



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

24-03-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	C	A	A	D	A	B	A	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	D	C	B	A	B	C	A	D

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	D	B	A	B	C	D	C	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	D	D	D	B	A	D	D	C

PART-3 : MATHEMATICS

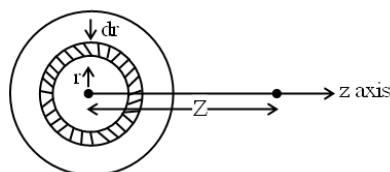
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	C	B	A	D	D	A	D	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	C	A	B	C	B	D	C	D

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (A)**
Consider a small ring of radius r and thickness dr on disc.



area of elemental ring on disc

$$dA = 2\pi r dr$$

charge on this ring

$$dq = \sigma dA$$

$$dE_z = \frac{k dq z}{(z^2 + r^2)^{3/2}}$$

$$E = \int_0^R dE_z = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R^2 + z^2}} \right]$$

2. **Ans (B)**

$$\frac{N}{N_0} = \left(\frac{1}{2} \right)^{\frac{t}{t_{1/2}}}$$

$$\frac{N}{10^{10}} = \left(\frac{1}{2} \right)^{\frac{30}{60}}$$

$$\Rightarrow N = 10^{10} \times \left(\frac{1}{2} \right)^{\frac{1}{2}} = \frac{10^{10}}{\sqrt{2}} \approx 7 \times 10^9$$

3. **Ans (C)**

$$[\mu_r] = 1 \text{ as } \mu_r = \frac{\mu}{\mu_m}$$

$$[\text{power factor } (\cos \phi)] = 1$$

$$\mu_0 = \frac{B_0}{H} \text{ (unit = } \text{NA}^{-2} \text{) : Not dimensionless}$$

$$[\mu_0] = [\text{MLT}^{-2} \text{A}^{-2}]$$

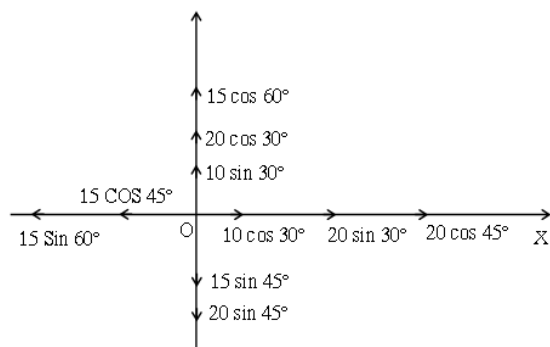
$$\text{quality factor (Q)} = \frac{\text{Energy stored}}{\text{Energy dissipated per cycle}}$$

So Q is unitless & dimensionless.

4. **Ans (A)**

Unit of $\frac{E}{H}$ is $\frac{\text{volt/metre}}{\text{Ampere/metre}} = \frac{\text{volt}}{\text{Ampere}} = \text{ohm}$

5. **Ans (A)**



$$\vec{F}_x = \left(10 \times \frac{\sqrt{3}}{2} + 20 \left(\frac{1}{2} \right) + 20 \left(\frac{1}{\sqrt{2}} \right) - 15 \left(\frac{1}{\sqrt{2}} \right) - 15 \left(\frac{\sqrt{3}}{2} \right) \right) \hat{i} = 9.25 \hat{i}$$

$$\vec{F}_y = \left(15 \left(\frac{1}{2} \right) + 20 \left(\frac{\sqrt{3}}{2} \right) + 10 \left(\frac{1}{2} \right) - 15 \left(\frac{1}{\sqrt{2}} \right) - 20 \left(\frac{1}{\sqrt{2}} \right) \right) \hat{j} = 5 \hat{j}$$

6. **Ans (D)**

$$P_m = \rho RT$$

$$\therefore \frac{P_1}{P_2} = \frac{\rho_1 T_1}{\rho_2 T_2}$$

$$\frac{\rho_1}{\rho_2} \Rightarrow \frac{P_1 T_2}{P_2 T_1} = \left(\frac{76}{45} \right) \times \frac{266}{300}$$

$$\frac{\rho_1}{\rho_2} \Rightarrow \frac{M_1}{M_2} = \frac{76 \times 266}{45 \times 300}$$

$$\therefore M_2 \Rightarrow \frac{45 \times 300 \times 185}{76 \times 266} = 123.54 \text{ kg}$$

7. **Ans (A)**

Using Newton's formula

$$(f + d_1)(f - d_2) = f^2$$

$$f^2 + fd_1 - fd_2 - d_1 d_2 = f^2$$

$$f = \frac{d_1 d_2}{d_1 - d_2}$$

$$\therefore R = \frac{2d_1 d_2}{d_1 - d_2}$$

8. **Ans (B)**

$$R = \frac{\ell}{\theta}$$

$$\text{Time} = \frac{4 \times 2\pi R}{v} = \frac{4 \times 2\pi}{v} \left(\frac{\ell}{\theta} \right)$$

$$\text{put } \ell = 4.4 \times 9.46 \times 10^{15}$$

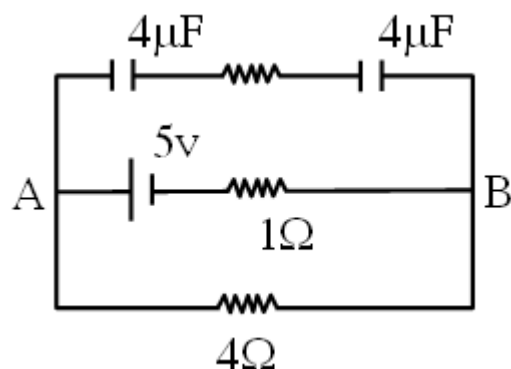
$$v = 8 \times 1.5 \times 10^{11}$$

$$\theta = \frac{4}{3600} \times \frac{\pi}{180} \text{ rad.}$$

$$\text{we get time} = 4.5 \times 10^{10} \text{ sec}$$

9. **Ans (A)**

On simplifying circuit we get



No current in upper wire.

$$\therefore V_{AB} =$$

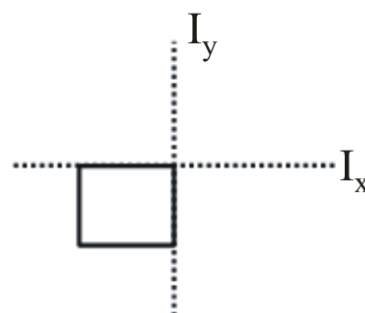
$$\frac{5}{4+1} \times 4 = 4 \text{ v.}$$

$$\therefore \theta = (C_{eq})v$$

$$\Rightarrow 2 \times 4 = 8 \mu\text{C}$$

10. **Ans (D)**

According to perpendicular Axis theorem.



$$I_x + I_y = I_z$$

$$I_z \Rightarrow \frac{m\ell^2}{3} + \frac{m\ell^2}{3} = \frac{2m\ell^2}{3}$$

11. **Ans (D)**

Active region of the CE transistor is linear region and is best suited for its use as an amplifier

12. **Ans (C)**

$$PT^3 = \text{constant} \left(\frac{nRT}{V} \right) T^3 = \text{constant}$$

$$T^4 V^{-1} = \text{constant}$$

$$T^4 = kV \Rightarrow 4 \frac{\Delta T}{T} = \frac{\Delta V}{V} \dots \dots \dots (1)$$

$$\Delta V = V \gamma \Delta T \dots \dots \dots (2)$$

comparing (1) and (2)

we get

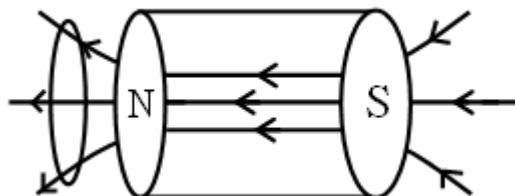
$$\gamma = \frac{4}{T}$$

13. **Ans (D)**

→ Increasing intensity means number of incident photons are increased.

→ Kinetic energy of ejected electrons depend on the frequency of incident photons, not the intensity.

14. **Ans (C)**



→ When magnet passes through centre region of solenoid, no current / Emf is induced in loop.

→ While entering flux increases so negative induced emf

→ While leaving flux decreases so positive induced emf.

15. **Ans (B)**

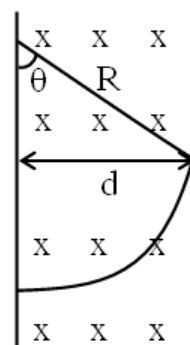
$$r = \frac{P}{qB} = \frac{\sqrt{2mk}}{qB}$$

Given they have same kinetic energy

$$r \propto \frac{\sqrt{m}}{q}$$

$$\frac{r_1}{r_2} = \frac{\sqrt{4}}{2} \times \frac{3}{\sqrt{16}} = \frac{3}{4}$$

$$r_2 = \frac{4r_1}{3} \quad (r_2 \text{ is for heavier ion and } r_1 \text{ is for lighter ion})$$



$$\sin \theta = \frac{d}{R}$$

$\theta \rightarrow$ Deflection

$$\theta \propto \frac{1}{R}$$

($R \rightarrow$ Radius of path)

$$\therefore R_2 > R_1 \Rightarrow \theta_2 < \theta_1$$

16. **Ans (A)**

$$\frac{1}{V_1} + \frac{1}{30} = \frac{1}{10}$$

$$\frac{1}{V_1} = \frac{2}{30} \Rightarrow V_1 = 15 \text{ cm}$$

$$\frac{1}{V_2} - \frac{1}{10} = -\frac{1}{10}$$

$$\frac{1}{V_2} = 0 \quad V_2 = \infty$$

$$V_3 = 30 \text{ cm}$$

$$OV_3 = 75 \text{ cm}$$

17. **Ans (B)**

Viscous force = Weight

$$= \rho \times \left(\frac{4}{3} \pi r^3 \right) g$$

$$= 3.9 \times 10^{-10}$$

18. Ans (C)

$$V = \frac{\omega}{K} = \frac{10 \times 10^{10}}{500} = 2 \times 10^8$$

$$V = \frac{2C}{3}$$

19. Ans (A)

$$i_1 = \frac{25}{5 + R}$$

$$i_2 = \frac{5}{R + \frac{1}{5}}$$

$$i_1 = i_2 \Rightarrow 5 \left(R + \frac{1}{5} \right) = 5 + R$$

$$4R = 4$$

$$R = 1 \Omega$$

20. Ans (D)

Potential energy is maximum at maximum distance from mean.

PART-1 : PHYSICS

SECTION-II

1. Ans (2)

Energy is maximum when mass is split equally so

$$\frac{M}{m} = 2$$

2. Ans (11)

$$f_{rms}^2 = f_{1rms}^2 + f_{2rms}^2$$

$$= \left(\frac{\sqrt{42}}{\sqrt{2}} \right)^2 + 10^2$$

$$= 121 \Rightarrow f_{rms} = 11 \text{ A}$$

3. Ans (3)

$$\text{In triangle shape } N_t = \frac{24a}{3a} = 8 \text{ In square}$$

$$N_s = \frac{24a}{4a} = 6 \quad \frac{M_t}{M_s} = \frac{N_t I A_t}{N_s I A_s} \quad [I \text{ will}$$

$$\text{be same in both}] = \frac{8 \times \frac{\sqrt{3}}{4} \times a^2}{6 \times a^2}$$

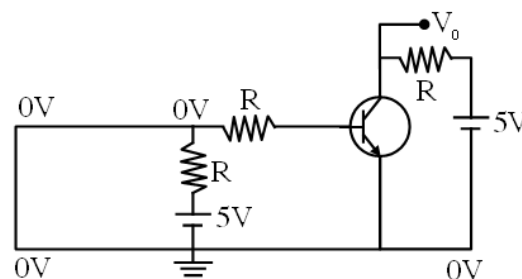
$$\frac{M_t}{M_s} = \frac{1}{\sqrt{3}}$$

$$y = 3$$

4. Ans (5)

As diodes D_1 and D_2 are in forward bias, so they acted as negligible resistances

$\Rightarrow$ Input voltage become zero



$\Rightarrow$ Input current is zero

$\Rightarrow$ Output current is zero

$\Rightarrow V_0 = 5 \text{ volt}$

5. Ans (20)

In series

$$R_{eq} = nR = 10 \text{ n}$$

$$i_s = \frac{20}{10 + 10n} = \frac{2}{1 + n} \text{ in parallel}$$

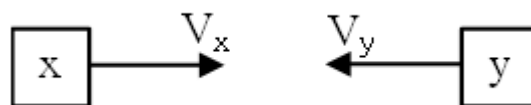
$$R_{eq} = \frac{10}{n}$$

$$i_p = \frac{20}{\frac{10}{n} + 10} = \frac{2n}{1 + n}$$

$$\frac{i_p}{i_s} = 20 \frac{\left(\frac{2n}{1+n} \right)}{\left(\frac{2}{1+n} \right)} = 20$$

$$n = 20$$

6. Ans (1210)



$$V_x = 36 \text{ km/hr} = 10 \text{ m/s}$$

$$V_y = 72 \text{ km/hr} = 20 \text{ m/s}$$

$$\text{by doppler's effect } F' = F_0 \left(\frac{V \pm V_0}{V \pm V_s} \right)$$

$$1320 = F_0 \left(\frac{340 + 20}{340 - 10} \right)$$

$$\Rightarrow F_0 = 1210 \text{ Hz}$$

7. **Ans (12)**

$$V = \sqrt{5000 + 24x}$$

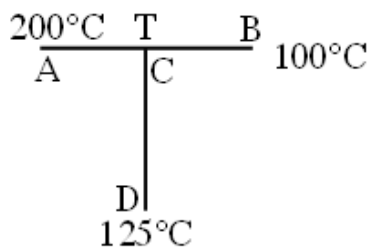
$$\frac{dV}{dx} = \frac{1}{2\sqrt{5000 + 24x}} \times 24 = \frac{12}{\sqrt{5000 + 24x}}$$

$$\text{now } a = V \frac{dV}{dx}$$

$$= \sqrt{5000 + 24x} \times \frac{12}{\sqrt{5000 + 24x}}$$

$$a = 12 \text{ m/s}^2$$

8. **Ans (2)**



Rods are identical so

$$R_{AB} = R_{CD} = 10 \text{ KW}^{-1}$$

C is mid-point of AB, so

$$R_{AC} = R_{CB} = 5 \text{ KW}^{-1}$$

at point C

$$\frac{200 - T}{5} = \frac{T - 125}{10} + \frac{T - 100}{5}$$

$$2(200 - T) = T - 125 + 2(T - 100)$$

$$400 - 2T = T - 125 + 2T - 200$$

$$T = \frac{725}{5} = 145^\circ\text{C}$$

$$I_h = \frac{145 - 125}{10} w = \frac{20}{10} w$$

$$I_h = 2w$$

9. **Ans (2)**

$$\text{Given } W_A = W_B$$

$$F_A \cos 45^\circ = F_B \cos 60^\circ \quad F_A \times \frac{1}{\sqrt{2}} = F_B \times \frac{1}{2}$$

$$\frac{F_A}{F_B} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$x = 2$$

10. **Ans (224)**

$$d_m = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

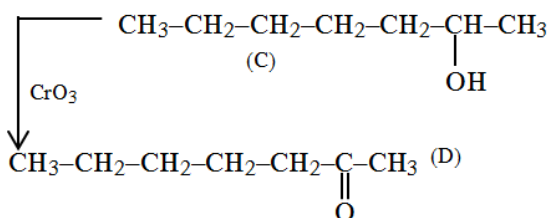
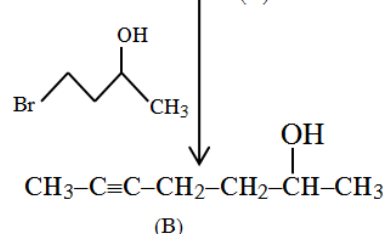
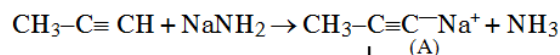
$$d_m = \left(\sqrt{2 \times 6400 \times 10^3 \times 320} + \sqrt{2 \times 6400 \times 10^3 \times 2000} \right) \text{ m}$$

$$d_m = 224 \text{ km}$$

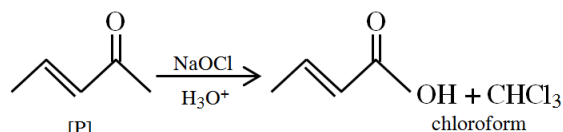
PART-2 : CHEMISTRY

SECTION-I

1. **Ans (D)**



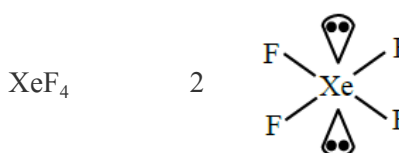
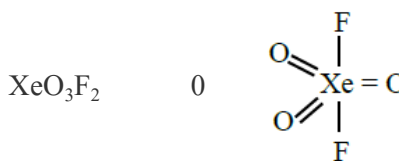
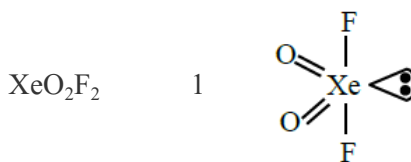
2. **Ans (A)**



NaOCl is used in haloform reaction as reagent.

3. **Ans (D)**

Species (Number of lone pairs of electrons on the central atom)



4. **Ans (B)**

Type of back bonding

BF_3 BCl BBr_3 BI_3
 $(2p\pi-2p\pi)$ $(2p\pi-3p\pi)$ $(2p\pi-4p\pi)$ $(2p\pi-5p\pi)$
 Therefore back bonding strength is as follows
 $\text{BF}_3 > \text{BCl} > \text{BBr}_3 > \text{BI}_3$

5. **Ans (A)**

The bond dissociation energy of D_2 is greater than H_2 and therefore D_2 reacts slower than H_2 .

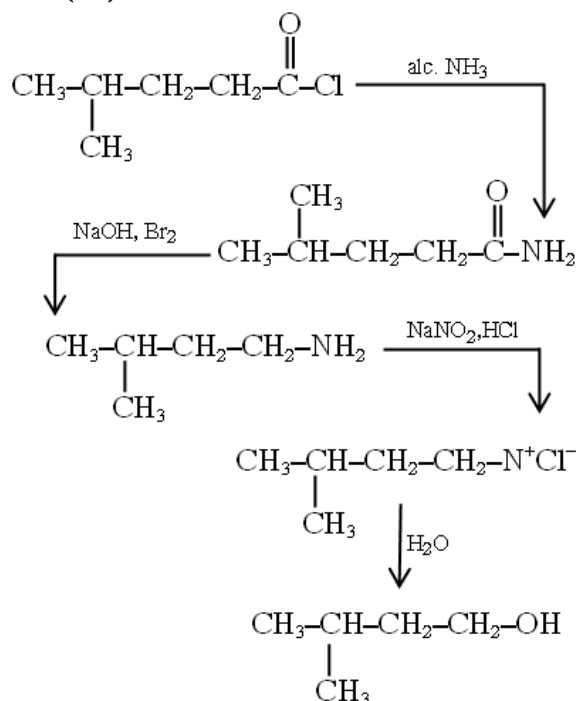
6. **Ans (B)**

Liquation method is used to purify those impure metals which has lower melting point than the melting point of impurities associated.
 $\therefore$ This method is used for metal having low melting point.

8. **Ans (D)**

B and C are tranquilizers

9. **Ans (C)**



10. **Ans (A)**

The intermolecular association is more prominent in case of primary amines as compared to secondary, due to the availability of two hydrogen atom.

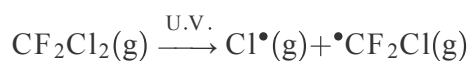
11. **Ans (D)**

$\text{FeCl}_3 + \text{K}_4[\text{Fe}(\text{CN})_6]$ (excess)

$\downarrow$
 $\text{K Fe}[\text{Fe}(\text{CN})_6]$
 Colloidal species

12. **Ans (D)**

In stratosphere CFCs get broken down by powerful UV radiations releasing $\text{Cl}^\bullet$



13. **Ans (D)**

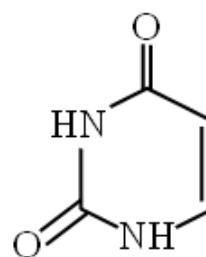
Gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
 Plaster of Paris $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
 Dead burnt plaster CaSO_4

14. **Ans (D)**

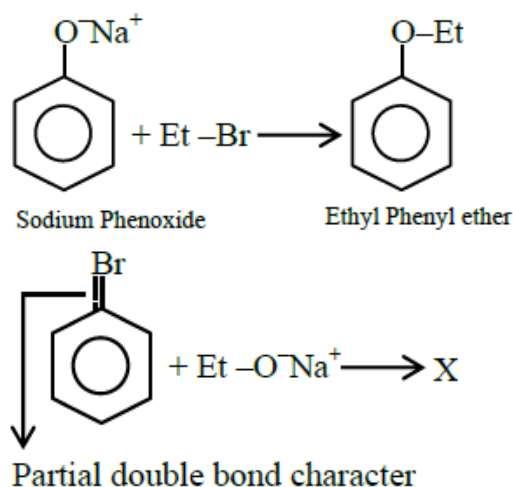
V_2O_3 basic
 CrO basic

15. **Ans (D)**

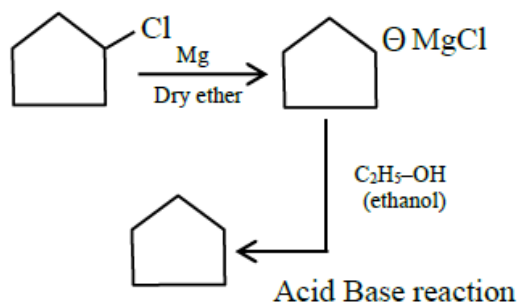
Isomeric form of uracil present in RNA



16. **Ans (B)**



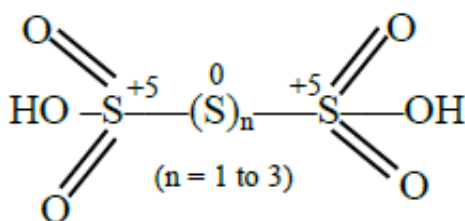
17. Ans (A)



18. Ans (D)

$$\frac{an^2}{V^2} = atm \Rightarrow a = atm \times \frac{dm^6}{mol^2}$$

19. Ans (D)



20. Ans (C)

Tyndall effect is observed in lyophobic colloids

PART-2 : CHEMISTRY

SECTION-II

1. Ans (40)

$$n_{AgBr} = \frac{0.188g}{188g/mol} = 10^{-3}mol$$

$$\Rightarrow n_{Br} = n_{AgBr} = 0.001 mol$$

$$\Rightarrow mass_{Br} = (0.001 \times 80) gm = 0.08 gm$$

$$\Rightarrow mass \% = \frac{0.08 \times 100}{0.2} = 40\%$$

2. Ans (4)

$$\left(\frac{\text{Rate of disappearance of } C_2H_6O}{3} \right) = \left(\frac{\text{Rate of appearance of } Cr_2(SO_4)_3}{2} \right)$$

$$\Rightarrow \left(\frac{2.67mol/min \times 3}{2} \right) = \text{rate of disappearance of } C_2H_6O.$$

$$\Rightarrow \text{Rate of disappearance of } C_2H_6O = 4.005 mol/min$$

3. Ans (3155)

$$mvr = \frac{nh}{2\pi} K.E. = \frac{n^2 h^2}{8\pi^2 mr^2} = \frac{4h^2}{8\pi^2 m(4a_0)^2}$$

$$= \left(\frac{4}{8\pi^2 \times 16} \right) \frac{h^2}{ma_0^2}$$

$$\Rightarrow x = 315.507$$

$$\Rightarrow 10x = 3155 \text{ (nearest integer)}$$

4. Ans (518)

Let mass of water initially present = x gm

$$\Rightarrow \text{Mass of sucrose} = (1000 - x) gm$$

$$\Rightarrow \text{moles of sucrose} = \left(\frac{1000 - x}{342} \right)$$

$$\Rightarrow 0.75 = \frac{\left(\frac{1000-x}{342} \right)}{\left(\frac{x}{1000} \right)} \Rightarrow \frac{x}{1000} = \frac{1000 - x}{342 \times 0.75}$$

$$\Rightarrow 256.5 x = 10^6 - 1000x$$

$$\Rightarrow x = 795.86 gm$$

$$\Rightarrow \text{moles of sucrose} = 0.5969$$

$$\Rightarrow \text{New mass of } H_2O = a kg$$

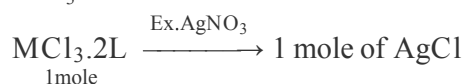
$$\Rightarrow 4 = \frac{0.5969}{a} \times 1.86$$

$$\Rightarrow a = 0.2775 kg$$

$$\Rightarrow \text{ice separated} = (795.86 - 277.5) = 518.3 gm$$

5. Ans (2)

$MCl_3 \cdot 2L$ octahedral



Its means that one Cl^- ion present in ionization sphere.

$$\therefore \text{formula} = [MCl_2L_2]Cl$$

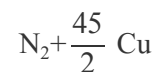
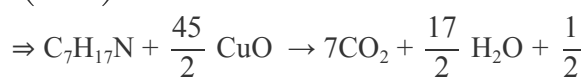
For octahedral complex coordination no. is 6

$\therefore L$ act as bidentate ligand

6. Ans (1125)

Moles of N in N,N - dimethylaminopentane =

$$\left(\frac{57.5}{115} \right) = 0.5mol$$



$$\frac{n_{CuO \text{ reacted}}}{\left(\frac{45}{2} \right)} = \frac{n_{C_7H_{17}N \text{ reacted}}}{1}$$

$$\Rightarrow n_{CuO \text{ reacted}} = \left(\frac{45}{2} \right) \times 0.5 = 11.25$$

7. **Ans (4)**

$$\text{Np} = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2$$

$$4f^{14} 5d^{10} 6p^6 7s^2 5f^4 6d^1$$

$$\text{Total no. of 'f' electron} = 14e^- + 4e^- = 18$$

8. **Ans (82)**

$$\Rightarrow \text{Millimoles of HCl} = 200 \times 0.2 = 40$$

$$\Rightarrow \text{Millimoles of NaOH} = 300 \times 0.1 = 30$$

$$\Rightarrow \text{Heat released} = \left(\frac{30}{1000} \times 57.1 \times 1000 \right) = 1713 \text{ J}$$

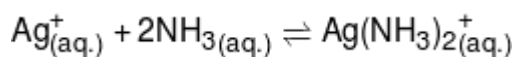
$$\Rightarrow \text{Mass of solution} = 500 \text{ ml} \times 1 \text{ gm/ml} = 500 \text{ gm}$$

$$\Rightarrow \Delta T = \frac{q}{m \times C} = \frac{1713 \text{ J}}{500 \text{ g} \times 4.18 \frac{\text{J}}{\text{g-K}}} = 0.8196 \text{ K}$$

$$= 81.96 \times 10^{-2} \text{ K}$$

9. **Ans (4)**

$$\text{Let moles added} = a$$



$$t = 0 \quad 0.8 \quad \left(\frac{a}{2} \right)$$

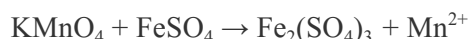
$$t = \infty \quad 5 \times 10^{-8} \quad \left(\frac{a}{2} - 1.6 \right) \quad 0.8$$

$$\frac{0.8}{(5 \times 10^{-8}) \left(\frac{a}{2} - 1.6 \right)^2} = 10^8$$

$$\Rightarrow \frac{a}{2} - 1.6 = 0.4 \Rightarrow a = 4$$

10. **Ans (316)**

$$\text{Let molarity of KMnO}_4 = x$$



$$n = 5 \quad n = 1$$

$$(\text{Equivalents of KMnO}_4 \text{ reacted}) = (\text{Equivalents of FeSO}_4 \text{ reacted})$$

$$\Rightarrow 5 \times x \times 10 \text{ ml} = 1 \times 0.1 \times 10 \text{ ml}$$

$$\Rightarrow x = 0.02 \text{ M}$$

$$\text{Molar mass of KMnO}_4 = 158 \text{ gm/mol}$$

$$\Rightarrow \text{Strength} = (x \times 158) = 3.16 \text{ g/l}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

$$\text{Let } t = \frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots \infty$$

$$= \left(2 - \frac{1}{2} \right) x^2 + \left(2 - \frac{1}{3} \right) x^3 + \left(2 - \frac{1}{4} \right) x^4 + \dots \infty$$

$$= 2(x^2 + x^3 + x^4 + \dots \infty) - \left(\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty \right)$$

$$= \frac{2x^2}{1-x} - (\ln(1-x) - x)$$

$$\Rightarrow t = \frac{2x^2}{1-x} + x - \ln(1-x)$$

$$\Rightarrow t = \frac{x(1+x)}{1-x} - \ln(1-x)$$

2. **Ans (D)**

$$\frac{2(1+2+3+\dots+y)}{3(1+2+3+\dots+y)} = \frac{4}{\log_{10} x}$$

$$\Rightarrow \log_{10} x = 6 \Rightarrow x = 10^6$$

Now,

$$y = (\log_{10} x) + \left(\log_{10} x^{\frac{1}{3}} \right) + \left(\log_{10} x^{\frac{1}{9}} \right) + \dots \infty$$

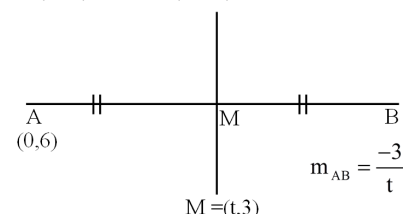
$$= \left(1 + \frac{1}{3} + \frac{1}{9} + \dots \infty \right) \log_{10} x$$

$$= \left(\frac{1}{1 - \frac{1}{3}} \right) \log_{10} x = 9$$

$$\text{So, } (x, y) = (10^6, 9)$$

3. **Ans (C)**

$$A(0,6) \text{ and } B(2t,0)$$



Perpendicular bisector of AB is

$$(y-3) = \frac{t}{3}(x-t)$$

$$\text{So, } C = \left(0, 3 - \frac{t^2}{3} \right)$$

Let P be (h, k)

$$h = \frac{t}{2}; k = \left(3 - \frac{t^2}{6} \right)$$

$$\Rightarrow k = 3 - \frac{4h^2}{6} \Rightarrow 2x^2 + 3y - 9 = 0$$

4. Ans (B)

$$\begin{aligned} \text{Given } a &= (\sin^{-1}x)^2 - (\cos^{-1}x)^2 \\ &= (\sin^{-1}x + \cos^{-1}x)(\sin^{-1}x - \cos^{-1}x) \\ &= \frac{\pi}{2} \left(\frac{\pi}{2} - 2\cos^{-1}x \right) \\ \Rightarrow 2\cos^{-1}x &= \frac{\pi}{2} - \frac{2a}{\pi} \\ \Rightarrow \cos^{-1}(2x^2 - 1) &= \frac{\pi}{2} - \frac{2a}{\pi} \\ \Rightarrow 2x^2 - 1 &= \cos\left(\frac{\pi}{2} - \frac{2a}{\pi}\right) \end{aligned}$$

5. Ans (A)

$$\text{Given matrix } A = \begin{bmatrix} 0 & 2 \\ k & -1 \end{bmatrix}$$

$$A^4 + 3IA = 2I$$

$$\Rightarrow A^4 = 2I - 3A$$

Also characteristic equation of A is

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 0 - \lambda & 2 \\ k & -1 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda + \lambda^2 - 2k = 0$$

$$\Rightarrow A + A^2 = 2KI$$

$$\Rightarrow A^2 = 2KI - A$$

$$\Rightarrow A^4 = 4K^2I + A^2 - 4AK$$

$$\text{Put } A^2 = 2KI - A$$

$$\text{and } A^4 = 2I - 3A$$

$$2I - 3A = 4K^2I + 2KI - A - 4AK$$

$$\Rightarrow I(2 - 2K - 4K^2) = A(2 - 4K)$$

$$\Rightarrow -2I(2K^2 + K - 1) = 2A(1 - 2K)$$

$$\Rightarrow -2I(2K - 1)(K + 1) = 2A(1 - 2K)$$

$$\Rightarrow (2K - 1)(2A) - 2I(2K - 1)(K + 1) = 0$$

$$\Rightarrow (2K - 1)[2A - 2I(K + 1)] = 0$$

$$\Rightarrow K = \frac{1}{2}$$

6. Ans (D)

$$\begin{aligned} &A(1, -2, 3) \\ &\vec{r} = (1, -2, 3) + \lambda(2, 3 - 6) \\ &\boxed{(1 + 2\lambda, -2 + 3\lambda, 3 - 6\lambda)} \quad x - y + z = 5 \end{aligned}$$

$$(1 + 2\lambda) + 2 - 3\lambda + 3 - 6\lambda = 5$$

$$\Rightarrow 6 - 7\lambda = 5 \Rightarrow \lambda = \frac{1}{7}$$

$$\text{so, } P = \left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7} \right)$$

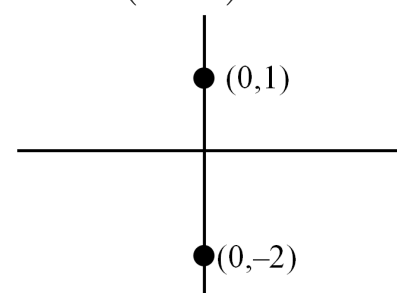
$$AP = \sqrt{\left(1 - \frac{9}{7}\right)^2 + \left(-2 + \frac{11}{7}\right)^2 + \left(3 - \frac{15}{7}\right)^2}$$

$$AP = \sqrt{\left(\frac{4}{49}\right) + \frac{9}{49} + \frac{36}{49}} = 1$$

7. Ans (D)

$$\text{Given } \frac{z - i}{z + 2i} \in \mathbb{R}$$

$$\text{Then } \arg\left(\frac{z - i}{z + 2i}\right) \text{ is } 0 \text{ or } \pi$$



$\Rightarrow S$ is straight line in complex

8. Ans (A)

$$\frac{dy}{dx} - 2xy = 2(2 \sin x - 5)x - 2 \cos x$$

$$\text{IF} = e^{-x^2}$$

$$\text{so, } y \cdot e^{-x^2} = \int e^{-x^2} (2x(2 \sin x - 5) - 2 \cos x) dx$$

$$\Rightarrow y \cdot e^{-x^2} = e^{-x^2} (5 - 2 \sin x) + c$$

$$\Rightarrow y = 5 - 2 \sin x + c \cdot e^{x^2}$$

$$\text{Given at } x = 0, y = 7$$

$$\Rightarrow 7 = 5 + c \Rightarrow c = 2$$

$$\text{So, } y = 5 - 2 \sin x + 2e^{x^2}$$

$$\text{Now at } x = \pi,$$

$$y = 5 + 2e^{\pi^2}$$

9. **Ans (D)**

Required equation of plane

$$P_1 + \lambda P_2 = 0$$

$$(x - y - z - 1) + \lambda(2x + y - 3z + 4) = 0$$

Given that its dist. From origin is $\frac{2}{\sqrt{21}}$

Thus

$$\frac{|4\lambda - 1|}{\sqrt{(2\lambda + 1)^2 + (\lambda - 1)^2 + (-3\lambda - 1)^2}} = \frac{\sqrt{2}}{\sqrt{21}}$$

$$\Rightarrow 21(4\lambda - 1)^2 = 2(14\lambda^2 + 8\lambda + 3)$$

$$\Rightarrow 336\lambda^2 - 168\lambda + 21 = 28\lambda^2 + 16\lambda + 6$$

$$\Rightarrow 308\lambda^2 - 184\lambda + 15 = 0$$

$$\Rightarrow 308\lambda^2 - 154\lambda - 30\lambda + 15 = 0$$

$$\Rightarrow (2\lambda - 1)(154\lambda - 15) = 0$$

$$\Rightarrow \lambda = \frac{1}{2} \text{ or } \frac{15}{154}$$

for $\lambda = \frac{1}{2}$ reqd. plane is

$$4x - y - 5z + 2 = 0$$

10. **Ans (A)**

$$U_n = \prod_{r=1}^n \left(1 + \frac{r^2}{n^2}\right)^r$$

$$L = \lim_{n \rightarrow \infty} (U_n)^{-4/n^2}$$

$$\log L = \lim_{n \rightarrow \infty} \frac{-4}{n^2} \sum_{r=1}^n \log \left(1 + \frac{r^2}{n^2}\right)^r$$

$$\Rightarrow \log L = \lim_{n \rightarrow \infty} \sum_{r=1}^n -\frac{4r}{n} \cdot \frac{1}{n} \log \left(1 + \frac{r^2}{n^2}\right)$$

$$\Rightarrow \log L \Rightarrow -4 \int_0^1 x \log(1 + x^2) dx$$

$$\text{put } 1 + x^2 = t$$

$$\text{Now, } 2x dx = dt$$

$$= -2 \int_1^2 \log(t) dt = -2[t \log t - t]_1^2$$

$$\Rightarrow \log L = -2(2 \log 2 - 1)$$

$$\therefore L = e^{-2(2 \log 2 - 1)}$$

$$= e^{-2(\log(\frac{4}{e}))}$$

$$= e^{\log(\frac{4}{e})^{-2}}$$

$$= \left(\frac{e}{4}\right)^2 = \frac{e^2}{16}$$

11. **Ans (A)**

$$(p \wedge (p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow r$$

$$\equiv (p \wedge (\sim p \vee q) \vee (\sim q \vee r)) \rightarrow r$$

$$\equiv ((p \wedge q) \wedge (\sim p \vee r)) \rightarrow r$$

$$\equiv (p \wedge q \wedge r) \rightarrow r$$

$$\equiv \sim (p \wedge q \wedge r) \vee r$$

$$\equiv (\sim p) \vee (\sim q) \vee (\sim r) \vee r$$

$$\Rightarrow \text{tautology}$$

12. **Ans (C)**

$$y + \frac{xdy}{dx} = x^2 \quad (\text{given})$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = x$$

$$\text{If } = e^{\int \frac{1}{x} dx} = x$$

Solution of DE

$$\Rightarrow y \cdot x = \int x \cdot x dx$$

$$\Rightarrow xy = \frac{x^3}{3} + \frac{c}{3}$$

Passes through $(-2, 2)$, so

$$-12 = -8 + c \Rightarrow c = -4$$

$$\therefore 3xy = x^3 - 4$$

$$\text{ie. } 3x \cdot f(x) = x^3 - 4$$

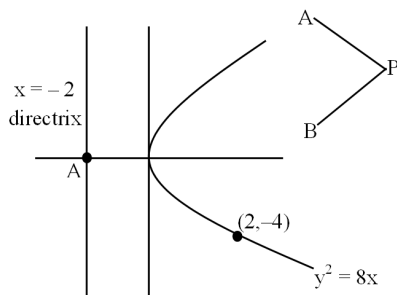
13. **Ans (C)**

$$\sum_{k=0}^{20} {}^{20}C_k \cdot {}^{20}C_{20-k}$$

sum of suffix is const. so summation will be

$${}^{40}C_{20}$$

14. Ans (A)



Equation of tangent at (2, -4) (T = 0)

$$-4y = 4(x + 2)$$

$$x + y + 2 = 0 \quad \dots(1)$$

equation of normal

$$x - y + \lambda = 0$$

$$\downarrow (2, -4)$$

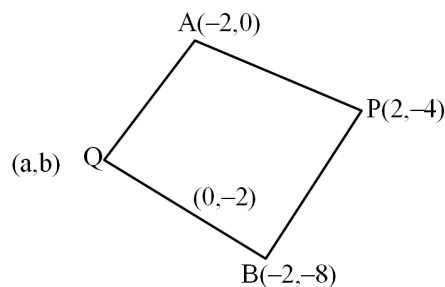
$$\lambda = -6$$

thus $x - y = 6 \dots(2)$ equation of normal

POI of (1) & $x = -2$ is A(-2, 0)

POI of (2) & $x = -2$ is A(-2, 8)

Given AQBP is a sq.



$$\Rightarrow m_{AQ} \cdot m_{AP} = -1$$

$$\Rightarrow \left(\frac{b}{a+2} \right) \left(\frac{4}{-4} \right) = -1 \Rightarrow a+2=b \quad \dots(1)$$

Also PQ must be parallel to x-axis thus

$$\Rightarrow b = -4$$

$$\therefore a = -6$$

$$\text{Thus } 2a + b = -16$$

15. Ans (B)

$$\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$$

As A, B, C are angles of triangle

$$A + B + C = \pi$$

$$A = \pi - (B + C)$$

$$\text{So, } \sin A = \sin(B + C) \quad \dots(1)$$

$$\text{Similarly } \sin B = \sin(A + C) \quad \dots(2)$$

From (1) and (2)

$$\frac{\sin(B+C)}{\sin(A+C)} = \frac{\sin(A-C)}{\sin(C-B)}$$

$$\sin(C+B) \cdot \sin(C-B) = \sin(A-C) \sin(A+C)$$

$$\sin^2 C - \sin^2 B = \sin^2 A - \sin^2 C$$

$$\{ \because \sin(x+y) \sin(x-y) = \sin^2 x - \sin^2 y \}$$

$$2\sin^2 C = \sin^2 A + \sin^2 B$$

By sine rule

$$2c^2 = a^2 + b^2$$

$$\Rightarrow b^2, c^2 \text{ and } a^2 \text{ are in A.P.}$$

16. Ans (C)

$$\begin{aligned} & \lim_{x \rightarrow \beta} \frac{e^{2(x^2+bx+c)} - 1 - 2(x^2+bx+c)}{(x-\beta)^2} \\ & \Rightarrow \lim_{x \rightarrow \beta} \frac{1 \left(1 + \frac{2(x^2+bx+c)}{1!} + \frac{2^2(x^2+bx+c)^2}{2!} + \dots \right) - 1 - 2(x^2+bx+c)}{(x-\beta)^2} \\ & \Rightarrow \lim_{x \rightarrow \beta} \frac{2(x^2+bx+c)}{(x-\beta)^2} \\ & \Rightarrow \lim_{x \rightarrow \beta} \frac{2(x-\alpha)^2(x-\beta)^2}{(x-\beta)^2} \\ & \Rightarrow 2(\beta-\alpha)^2 = 2(b^2-4c) \end{aligned}$$

17. **Ans (B)**

Probability of obtaining total sum 7 = probability of getting opposite faces.

Probability of getting opposite faces

$$= 2 \left[\left(\frac{1}{6} - x \right) \left(\frac{1}{6} + x \right) + \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} \right]$$

$$\Rightarrow 2 \left[\left(\frac{1}{6} - x \right) \left(\frac{1}{6} + x \right) + \frac{1}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{1}{6} \right] = \frac{13}{96}$$

(given)

$$x = \frac{1}{8}$$

18. **Ans (D)**

$$x^2 + 9y^2 - 4x + 3 = 0$$

$$(x^2 - 4x) + (9y^2) + 3 = 0$$

$$(x^2 - 4x + 4) + (9y^2) + 3 - 4 = 0$$

$$(x - 2)^2 + (3y)^2 = 1$$

$$\frac{(x - 2)^2}{(1)^2} + \frac{y^2}{\left(\frac{1}{3}\right)^2} = 1 \text{ (equation of an ellipse)}$$

As it is equation of an ellipse, x & y can vary inside the ellipse.

$$\text{So, } x - 2 \in [-1, 1] \text{ and } y \in \left[-\frac{1}{3}, \frac{1}{3}\right]$$

$$x \in [1, 3] \quad y \in \left[-\frac{1}{3}, \frac{1}{3}\right]$$

19. **Ans (C)**

$$\text{Let } I = \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$$

$$I = \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x - 22)^2} dx \dots (1)$$

we know

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx \text{ (king)}$$

$$\text{So } I = \int_6^{16} \frac{\log_e (22 - x)^2}{\log_e (22 - x)^2 + \log_e (22 - (22 - x))^2} dx$$

$$I = \int_0^{16} \frac{\log_e (22 - x)^2}{\log_e x^2 + \log_e (22 - x)^2} dx \dots (2)$$

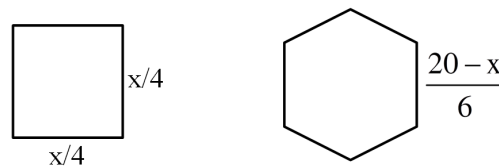
(1) + (2)

$$2I = \int_6^{16} 1 dx = 10$$

$$I = 5$$

20. **Ans (D)**

Let the wire is cut into two pieces of length x and 20 - x.



Area of square = $\left(\frac{x}{4}\right)^2$ Area of regular hexagon

$$= 6 \times \frac{\sqrt{3}}{4} \left(\frac{20 - x}{6}\right)^2$$

$$\text{Total area} = A(x) = \frac{x^2}{16} + \frac{3\sqrt{3}}{2} \frac{(20 - x)^2}{36}$$

$$A'(x) = \frac{2x}{16} + \frac{3\sqrt{3} \times 2}{2 \times 36} (20 - x) (-1)$$

$$A'(x) = 0 \text{ at } x = \frac{40\sqrt{3}}{3 + 2\sqrt{3}}$$

Length of side of regular Hexagon

$$= \frac{1}{6} (20 - x)$$

$$= \frac{1}{6} \left(20 - \frac{4\sqrt{3}}{3 + 2\sqrt{3}} \right)$$

$$= \frac{10}{2 + 2\sqrt{3}}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (90)**

Since, $\vec{a} \cdot \vec{b} = 0$

$$1 + 15 + \alpha\beta = 0 \Rightarrow \alpha\beta = -16 \dots (1)$$

Also,

$$|\vec{b} \times \vec{c}|^2 = 75 \Rightarrow (10 + \beta^2) 14 - (5 - 3\beta)^2 = 75$$

$$\Rightarrow 5\beta^2 + 30\beta + 40 = 0$$

$$\Rightarrow \beta = -4, -2$$

$$\Rightarrow \alpha = 4, 8$$

$$\Rightarrow |\vec{a}|_{\max}^2 = (26 + \alpha^2)_{\max} = 90$$

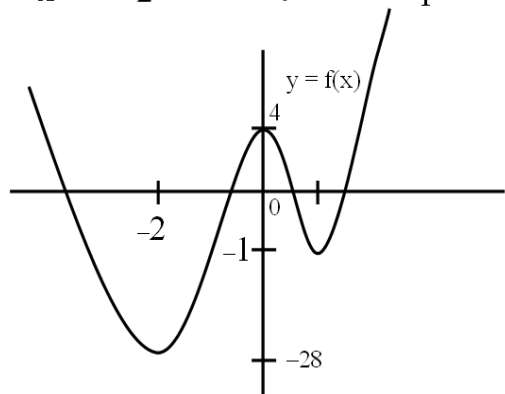
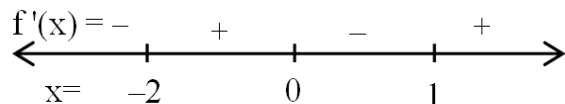
2. **Ans (4)**

$$3x^4 + 4x^3 - 12x^2 + 4 = 0$$

$$\text{So, let } f(x) = 3x^4 + 4x^3 - 12x^2 + 4$$

$$\therefore f'(x) = 12x(x^2 + x - 2)$$

$$= 12x(x+2)(x-1)$$



3. **Ans (61)**

$$r = \sqrt{\frac{p^2}{4} + \frac{(1-p)^2}{4}} - 5 = \frac{\sqrt{2p^2 - 2p - 19}}{2}$$

$$\text{Since, } r \in (0, 5]$$

$$\text{So, } 0 < 2p^2 - 2p - 19 \leq 100$$

$$\Rightarrow p \in \left[\frac{1-\sqrt{239}}{2}, \frac{1+\sqrt{39}}{2} \right) \cup \left(\frac{1+\sqrt{39}}{2}, \frac{1+\sqrt{239}}{2} \right]$$

so, number of integral values of p^2 is 61

4. **Ans (256)**

$$A = (-\infty, 1) \cup (3, \infty)$$

$$B = (-\infty, -2) \cup (2, \infty)$$

$$C = (-\infty, 2] \cup [6, \infty)$$

$$\text{So, } A \cap B \cap C = (-\infty, -2) \cup [6, \infty)$$

$$Z \cap (A \cap B \cap C)' = \{-2, -1, 0, -1, 2, 3, 4, 5\}$$

Hence no. of its subsets = $2^8 = 256$.

5. **Ans (15)**

$$I = \int \frac{dx}{\left[\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right]^2}$$

$$\int \frac{dt}{(t^2 + \frac{3}{4})^2} \quad \left(\text{Put } x + \frac{1}{2} = t \right)$$

$$= \frac{\sqrt{3}}{2} \int \frac{\sec^2 \theta d\theta}{\frac{9}{16} \sec^4 \theta} \quad \left(\text{Put } t = \frac{\sqrt{3}}{2} \tan \theta \right)$$

$$= \frac{4\sqrt{3}}{9} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{4\sqrt{3}}{9} \left[\theta + \frac{\sin 2\theta}{2} \right] + c$$

$$= \frac{4\sqrt{3}}{9} \left[\tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + \frac{\sqrt{3}(2x+1)}{3 + (2x+1)^2} \right] + c$$

$$= \frac{4\sqrt{3}}{9} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + \frac{1}{3} \left(\frac{2x+1}{x^2+x+1} \right) + c$$

$$\text{Hence, } 9(\sqrt{3}a + b) = 15$$

6. **Ans (5)**

$$2 \times (i) - (ii) - (iii) \text{ gives :}$$

$$-(1 + \beta)z = 3 - \alpha$$

For infinitely many solution

$$\beta + 1 = 0 = 3 - \alpha \Rightarrow (\alpha, \beta) = (3, -1)$$

$$\text{Hence, } \alpha + \beta - \alpha\beta = 5$$

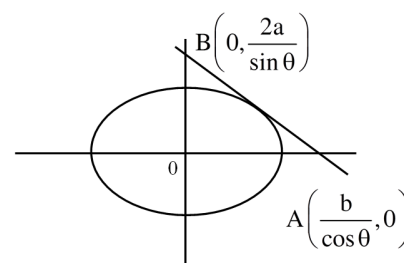
7. **Ans (13)**

$$\frac{n^2 - 1}{12} = 14 \Rightarrow n = 13$$

8. **Ans (2)**

Tangent

$$\frac{x \cos \theta}{b} + \frac{y \sin \theta}{2a} = 1$$



$$\text{So, area } (\triangle OAB) = \frac{1}{2} \times \frac{b}{\cos \theta} \times \frac{2a}{\sin \theta}$$

$$= \frac{2ab}{\sin 2\theta} \geq 2ab$$

$$\Rightarrow k = 2$$

9. **Ans (100)**

5	a	b	b	a	5
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It is always divisible by 5 and 11.

So, required number = $10 \times 10 = 100$

10. **Ans (17)**

$$y^{\frac{1}{4}} + \frac{1}{y^{\frac{1}{4}}} = 2x$$

$$\Rightarrow \left(y^{\frac{1}{4}}\right)^2 - 2xy^{\left(\frac{1}{4}\right)} + 1 = 0$$

$$\Rightarrow y^{\frac{1}{4}} = x + \sqrt{x^2 - 1} \text{ or } x - \sqrt{x^2 - 1}$$

$$\text{So, } \frac{1}{4} \frac{1}{y^{\frac{3}{4}}} \frac{dy}{dx} = 1 + \frac{x}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \frac{1}{4} \frac{1}{y^{\frac{3}{4}}} \frac{dy}{dx} = \frac{y^{\frac{1}{4}}}{\sqrt{x^2 - 1}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{4y}{\sqrt{x^2 - 1}} \quad \dots(1)$$

$$\text{Hence, } \frac{d^2y}{dx^2} = 4 \frac{\left(\sqrt{x^2 - 1}\right)' y' - \frac{yx}{\sqrt{x^2 - 1}}}{x^2 - 1}$$

$$\Rightarrow (x^2 - 1) y'' = 4 \frac{(x^2 - 1) y' - xy}{\sqrt{x^2 - 1}}$$

$$\Rightarrow (x^2 - 1) y'' = 4 \left(\sqrt{x^2 - 1} y' - \frac{xy}{\sqrt{x^2 - 1}} \right)$$

$$\Rightarrow (x^2 - 1) y'' = 4 \left(4y - \frac{xy'}{4} \right) \quad (\text{from 1})$$

$$\Rightarrow (x^2 - 1) y'' + xy' - 16y = 0$$

$$\text{So, } |\alpha - \beta| = 17$$